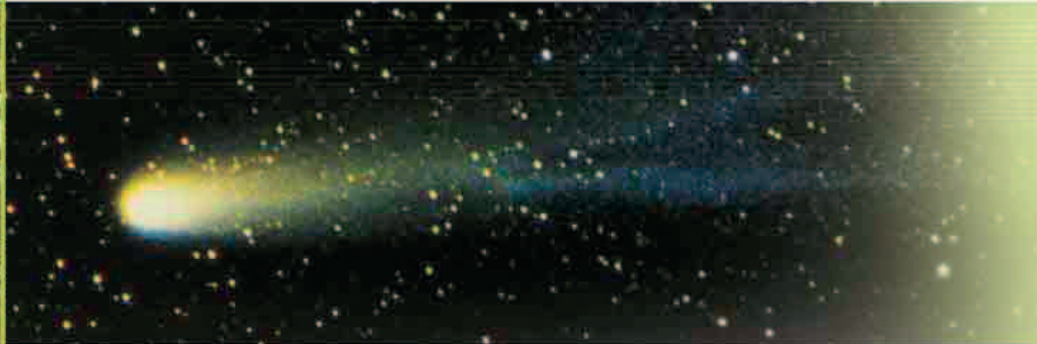




“ IN... 1456... A COMET WAS SEEN PASSING **RETROGRADE** BETWEEN THE **EARTH** AND THE **SUN**... HENCE I DARE VENTURE **TO FORETELL**, THAT IT WILL **RETURN** AGAIN IN... 1758. ”

Edmond Halley, from *A Synopsis of the Astronomy of Comets*, 1705

Edmond Halley correctly surmised that comets seen at regular 76-year intervals had been the same comet. It later became known as Comet Halley.

“ THE **CHANGING** OF BODIES INTO **LIGHT**, AND LIGHT INTO **BODIES**, IS VERY CONFORMABLE TO THE **COURSE OF NATURE**, WHICH SEEMS **DELIGHTED** WITH **TRANSMUTATIONS**. ”

Isaac Newton, from *Opticks*, 1704

ISAAC NEWTON PUBLISHED HIS SECOND GREAT SCIENTIFIC BOOK, entitled *Opticks*, in 1704. The experiments he described in this book proved that the spectrum of brilliant colors produced when sunlight shines through a prism is not an effect of the glass (see 1665–66). Instead, as Newton showed, the colors are all contained in “white” sunlight and are simply

separated when each color of light is bent, or refracted, differently as it enters the prism and slows down slightly. He also suggested that **light is a stream of tiny particles**, or “corpuscles,” traveling at great speed. The theory ignited a debate that lasted more than 200 years about whether light is indeed formed of particles or, as suggested by Newton’s Dutch rival Christiaan Huygens, waves.

That same year, English instrument maker and experimenter **Francis Hauksbee** (1660–1713) began a series of

demonstrations at the **Royal Society** in London on the effects of **static electricity**. In 1704, Hauksbee thrilled witnesses with a demonstration of “**barometric light**”—the sparks of light that appear when mercury in the vacuum at the top of a mercury barometer is shaken. Two years later, Hauksbee built the **first electrical machine**, which he called the “influence machine,” in which a hand-turned spindle rubbed wool against amber inside a glass vacuum globe to generate a glowing **static charge**. It was a forerunner of electric light.

In 1703, Dutch mathematician and astronomer **Christiaan Huygens** had published details of the gearing needed to drive a clockwork model of the Solar System that would precisely represent how the Sun and planets move in a year of 365.242 days. By 1704, English clockmakers **George Graham** (1764–51) and **Thomas Tompion** (1639–1713) had built a **clockwork mechanism**, based on Huygens’ calculation, to show how Earth and the Moon move around the Sun. The pair were asked to make another mechanism for English nobleman Charles Boyle, 4th Duke of Orrery. Such devices subsequently became known as orreries.



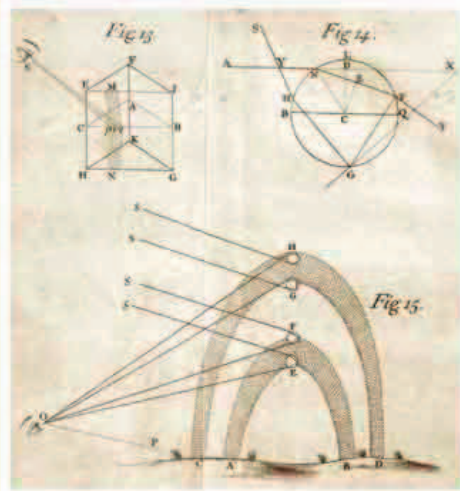
Earth and Moon

This orrery, made by George Graham and Thomas Tompion, shows how Earth and the Moon move around the Sun. Later orreries included the movement of all the planets.

to describe the ratio of the circumference of a circle to its diameter—approximately equal to the number 3.14159. Also this year, English inventor **Thomas Newcomen** (1663–1729) built a prototype for his steam engine that was to kickstart the Industrial Revolution in Europe (see 1712–13).

In 1705, English astronomer **Edmond Halley** explained how **comets are on a great elliptical journey** around the Sun, and appear periodically when their journey brings them near to the Sun and Earth. He argued that comets seen in 1456, 1531, 1607, and 1682 were a single comet—now known as Halley’s Comet—and predicted, correctly, that it would return in 1758.

In 1706, Welsh mathematician **William Jones** (1675–1749) proposed the **Greek letter pi (π)**



Splitting light
Isaac Newton’s findings, published in *Opticks* in 1704, showed that “white” sunlight contained all the colors of the rainbow.

1704 English astronomer, mathematician, and physicist **Isaac Newton** publishes *Opticks*

1704 English clockmakers **George Graham** and **Thomas Tompion** invent the orrery

1704 English scientist **Francis Hauksbee** demonstrates **barometric light**

1705 English astronomer **Edmond Halley** predicts the return of **Halley’s Comet**

1705 German naturalist **Maria Sibylla Merian** publishes an important study of butterflies

1706 English scientist **Francis Hauksbee** invents the static electric **influence machine**

1706 English ironmonger **Thomas Newcomen** invents a prototype of the steam engine